IT694 Computer Networks (Semester II, 2023-24)

Lectures Mon 12:00, Tue, Thu 10:00
Labs CEP 207

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Course Outline

The course explains the evolution of computer and communication networks and the design principles of modern network architectures. Primary focus is on system level concepts and engineering design and implementation issues. A top down approach is used to familiarize students with network application design and progressively define the underlying support needed to build such application. A thorough treatment of TCP/IP set of protocols is studied. In addition, we will study the design and implementation of modern network applications using sockets library.

The associated laboratory component is designed to expose students to the network simulation tools for learning the detailed working of the standard protocols. In addition, students will also work on developing network applications using the sockets library.

Course Outcomes

At the end of the course, students are expected to:

1. understand the TCP/IP Internet reference Model
2. be able to design and develop network applications
3. learn to measure performance of protocols using simulation tools

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X		X		X				X	X		X

Text Book

1. Kurose J.F. & Ross K.W., "Computer Networking - A Top-Down Approach Featuring the Internet," 7th edition or later, Pearson Education
2. W. Richard Stevens, "Unix Network Programming, Volume 1: The Sockets Networking API," 3rd edition, Addison-Wesley

Evaluation Scheme

1. Lab 30%
2. Mid Sem + Class Test(s) 30%
3. Final Exam 40%

Course Material

All the course material, including lecture notes, papers, and lab experiments will be maintained on **google classroom** for the course

Course Policies

1. It is expected that you attend all the lectures. **A minimum of 80% attendance is required. There will be a grade penalty for fewer than 80% attendance.**
2. You must perform **all** the lab experiments to get a passing grade.

Lecture Schedule

SlNo	Description		Lectures
1	Overview		3
	1.1	Overview	
	1.2	Internet-Birds' Eye View, History	
	1.3	Internet-Layered Architecture	
	1.4	Packet Switching, Best Effort Services	
2	Network Applications		8
	2.1	Client-Server Applications	
	2.2	Chat Application Design, Socket Programming	
	2.3	SFTP File Transfer Protocol	
	2.4	Domain Name Service, Mail, SMTP	
	2.5	Peer to Peer Search, Distributed Hash	
	2.6	Video Streaming, DASH	
3	End to End Issues		8
	5.1	Transport Layer Basics	
	5.2	Reliability	
	5.3	Connectionless and Connection Oriented Transport	
	5.4	TCP and UDP protocols	
	5.5	Congestion Management	
4	Routing and Addressing		8
	4.1	Introduction	
	4.2	Packet Switching, Batcher Banyan Switch	
	4.3	Routing - Introduction, Multicast, Broadcast	
	4.4	Addressing, CIDR, IP Protocol IPv4, IPv6	
	4.5	Hierarchical Routing, BGP, Mobile Routing	
	4.6	Control and Data Path, Open Flow	
5	Link Layer Technologies		5
	3.1	Introduction	
	3.2	Media Access Protocols, ALOHA	
	3.3	IEEE 802.3 Ethernet Protocol	
	3.4	Switched LAN, Virtual LANs	
6	Wireless Networks		5
	3.1	Introduction	
	3.2	IEEE 802.11 (WiFi) MAC protocol	
	3.3	Cellular architecture and Mobility management	

Lab Assignment Outline

Detailed assignments will be provided before each lab.

Network Applications using sockets (in C)

1. Introduction to sockets library functions (see the attached guide). Writing, and compiling a simple “echo” client and server for tcp and udp sockets.
2. Introduction to posix threads and converting the “echo” server to multithreaded server.
3. Design and implementation of an interesting “real-world” network application. It can be e.g. a group chat application with instant messenger features or some other application that can be completed over two weeks.

Understanding Standard TCP/IP protocols using NetSim simulator

1. Installation and GUI features of NetSim. Configuration of different type of networks and traffic profiles.
2. Measurement and Analysis. [will discuss the experiments later]

WireShark Experiments using Kurose & Ross textbook assignments.

1. HTTP, DNS, TCP protocol analysis